

NONG PHLUB AGROMET, ThailandWMO: **484740**Lat: **12.5893N**Lon: **99.7346E**Elev: **108**StdP: **100.04**Time Zone: **7.00 (E07)**Period: **08-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.8	18.3	11.7	8.7	21.3	13.4	9.7	22.4	6.9	29.0	6.2	28.7	0.2	N/A	0.293

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	11.1	36.8	25.3	35.7	25.4	34.8	25.4	27.1	32.7	26.8	32.2	26.6	31.9	2.7	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.8	21.4	29.2	25.5	21.0	28.9	25.2	20.6	28.6	86.3	32.9	84.9	32.3	83.6	32.1	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.9	5.2	4.6	14.6	38.6	1.8	1.2	13.3	39.4	12.3	40.1	11.3	40.8	10.0	41.7		
(5)			13.1	27.9	1.9	0.3	11.7	28.1	10.6	28.3	9.5	28.5	8.1	28.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.7	25.4	26.9	28.5	29.7	29.3	28.7	28.3	28.3	28.0	27.3	26.5	25.3		
	DBStd	1.98	1.96	1.57	1.67	1.62	1.29	1.11	1.06	1.04	0.98	1.05	1.27	1.97		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD10.0	6457	477	474	574	592	599	562	567	566	538	536	497	476		
	CDD18.3	3416	218	241	316	342	341	312	309	308	288	277	247	217		
	CDH23.3	33730	1876	2379	3481	3936	3665	3293	3149	3055	2661	2391	2068	1776		
	CDH26.7	14168	717	1098	1774	2025	1639	1333	1243	1193	990	803	722	631		
(14) Wind	WSAvg	0.9	1.1	1.0	1.1	0.9	0.7	0.8	0.7	0.7	0.6	0.6	0.9	1.2		
(15) Precipitation	PrecAvg	976	16	10	51	43	105	90	98	85	144	262	64	9		
	PrecMax	1385	69	75	421	131	356	164	173	175	297	548	314	41		
	PrecMin	693	0	0	0	0	28	33	28	36	84	88	0	0		
	PrecStd	171	19	18	85	35	65	33	37	29	41	127	74	11		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.6	35.2	37.5	38.7	37.2	35.6	35.0	34.9	34.6	33.3	33.5	33.2		
		MCWB	23.8	24.3	24.1	25.5	26.3	26.0	25.8	25.7	25.8	26.1	25.0	24.1		
	2%	DB	32.5	34.2	36.1	37.3	35.7	34.4	34.1	33.8	33.4	32.4	32.3	32.3		
		MCWB	23.4	24.1	24.4	25.5	26.3	25.9	25.6	25.5	25.8	25.9	24.7	23.9		
	5%	DB	31.6	33.2	35.0	36.1	34.6	33.4	33.1	32.9	32.5	31.7	31.5	31.3		
		MCWB	23.1	23.9	24.5	25.6	26.3	25.9	25.5	25.4	25.7	25.7	24.6	23.6		
	10%	DB	30.5	32.2	34.0	34.8	33.4	32.4	32.0	31.9	31.4	30.8	30.6	30.2		
		MCWB	22.8	23.8	24.5	25.5	26.2	25.8	25.4	25.3	25.5	25.6	24.5	23.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.6	25.8	26.5	27.4	27.7	27.3	26.9	26.7	27.1	27.0	26.5	25.7		
		MCDB	30.5	32.4	32.2	34.6	33.4	32.3	32.2	31.7	31.7	31.4	30.5	30.5		
	2%	WB	25.0	25.2	25.9	26.9	27.1	26.8	26.4	26.2	26.6	26.5	25.9	25.0		
		MCDB	30.0	31.6	32.3	33.8	32.9	31.9	31.7	31.2	31.2	30.7	30.1	30.1		
	5%	WB	24.5	24.8	25.5	26.5	26.7	26.5	26.0	26.0	26.2	26.1	25.6	24.5		
		MCDB	29.2	31.0	31.9	33.2	32.3	31.5	31.3	31.0	30.8	30.3	29.6	29.3		
	10%	WB	23.9	24.4	25.1	26.1	26.4	26.1	25.7	25.6	25.8	25.8	25.2	23.9		
		MCDB	28.3	30.4	31.5	32.6	31.8	31.0	30.6	30.4	30.2	29.6	29.0	28.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.6	11.4	11.8	11.1	9.2	8.0	7.9	8.1	8.0	7.2	7.7	9.5		
		MCDBR	11.8	12.6	13.3	12.9	10.8	9.6	9.6	9.7	9.5	8.6	9.6	10.7		
		MCWBR	3.9	3.8	3.7	3.3	3.0	2.8	2.8	2.7	3.1	3.0	3.2	3.5		
	5% WB	MCDBR	9.4	11.0	12.0	11.6	9.6	8.5	8.5	8.6	8.7	7.7	7.9	9.3		
		MCWBR	3.3	3.7	3.8	3.3	3.1	2.9	2.9	2.8	3.2	3.0	3.0	3.4		
(40) Clear-Sky Solar Irradiance	taub		0.512	0.555	0.571	0.518	0.458	0.438	0.436	0.437	0.440	0.518	0.506	0.493		
	taud		2.058	1.965	1.962	2.149	2.359	2.430	2.429	2.415	2.394	2.132	2.136	2.133		
	Ebn at Noon		790	773	770	807	842	849	853	863	868	794	791	796		
	Edh at Noon		165	187	191	158	126	115	116	119	122	156	151	150		
(44) All-Sky Solar Radiation	RadAvg		4.90	5.44	5.57	5.83	5.03	4.54	4.27	4.33	4.48	4.33	4.63	4.74		
	RadStd		0.34	0.28	0.47	0.39	0.49	0.36	0.37	0.39	0.35	0.35	0.41	0.30		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page